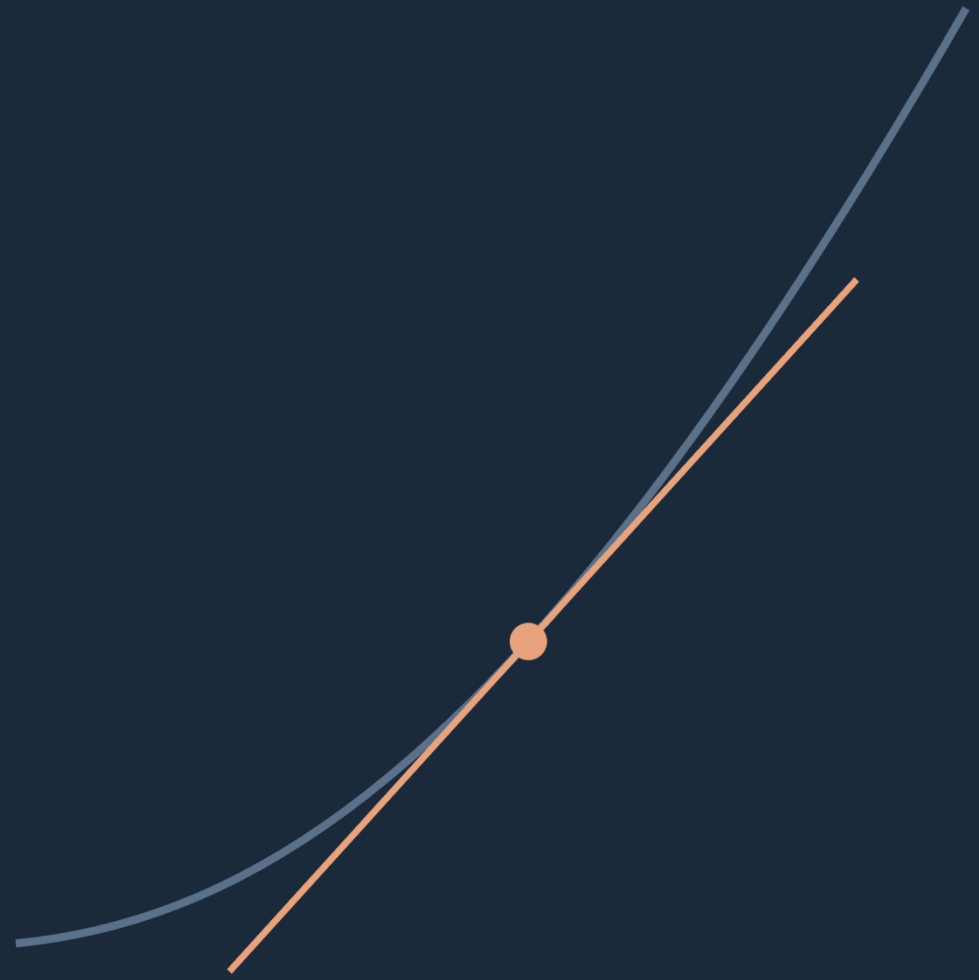
CALCULUS

Derivatives

From slopes to rates of change



Erisa Terolli

A QUESTION TO START

A runner passes you. How fast are they going at exactly t = 1 second?

Speed is distance divided by time, but at a single instant no time passes. The derivative is the tool that makes "speed at an instant" precise. We will answer this question with real data at the end of the lecture.

LEARNING OBJECTIVES

By the end of this lecture you can

01

Explain the derivative as the slope of a tangent line and as an instantaneous rate of change.

02

Compute a derivative directly from the limit definition.

03

Apply the power, product, quotient and chain rules, and differentiate the functions ML uses: exponentials, logs and the sigmoid.

04

Estimate a derivative from data, and use finite differences to check a derivative you computed.

BACKGROUND

What is calculus?

Calculus studies how quantities change (derivatives) and how they accumulate (integrals).

Two inventors

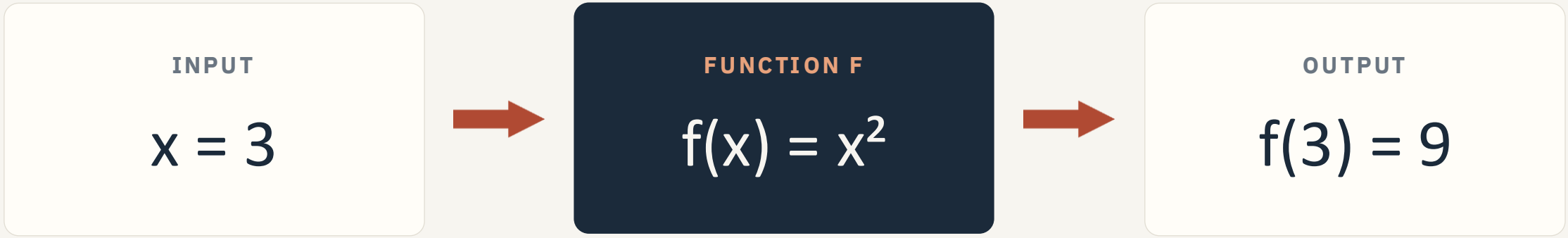
Isaac Newton and Gottfried Wilhelm Leibniz developed calculus independently in the late 1600s. The dy/dx notation we use today is Leibniz's.

Today

The "change" half: derivatives.

FUNCTIONS

A function turns an input into an output



Each input gives exactly one output. The derivative asks the next question: **when the input changes a little, how much does the output change?**

SLOPE

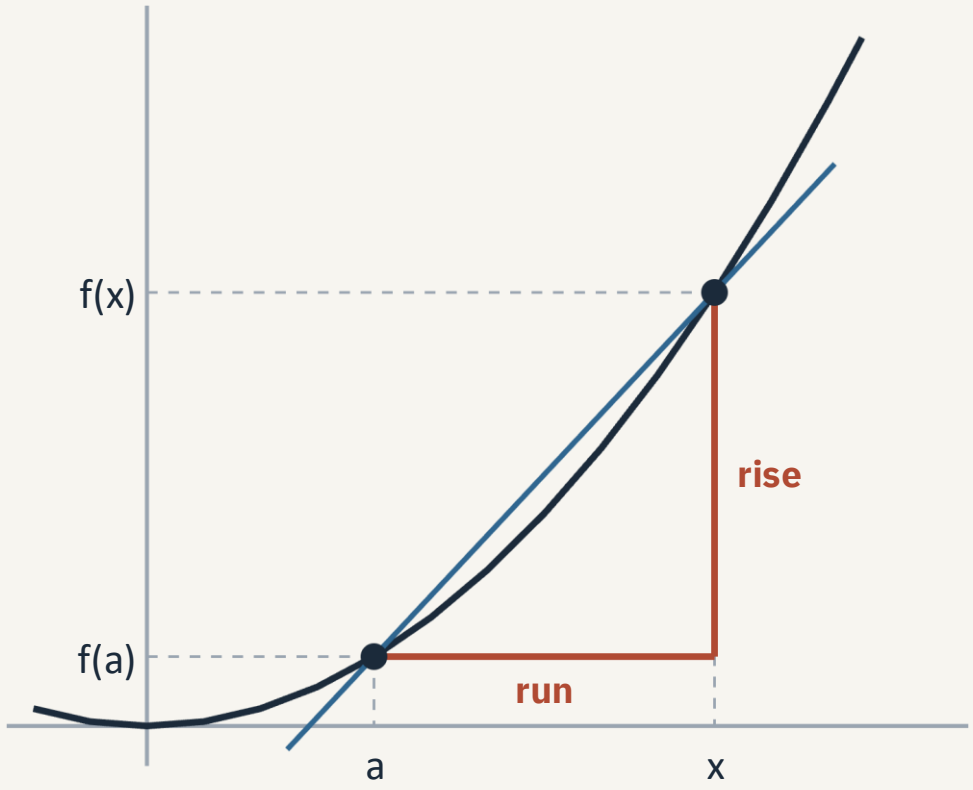
Slope measures direction and steepness

The slope of the line through two points on a graph is rise over run:

$$m = \frac{\text{rise}}{\text{run}} = \frac{f(x) - f(a)}{x - a}$$

Write the second point as $x = a + h$. Then the run is just h:

$$m = \frac{f(a + h) - f(a)}{h}$$



RATE OF CHANGE

Shrink the gap: the tangent line

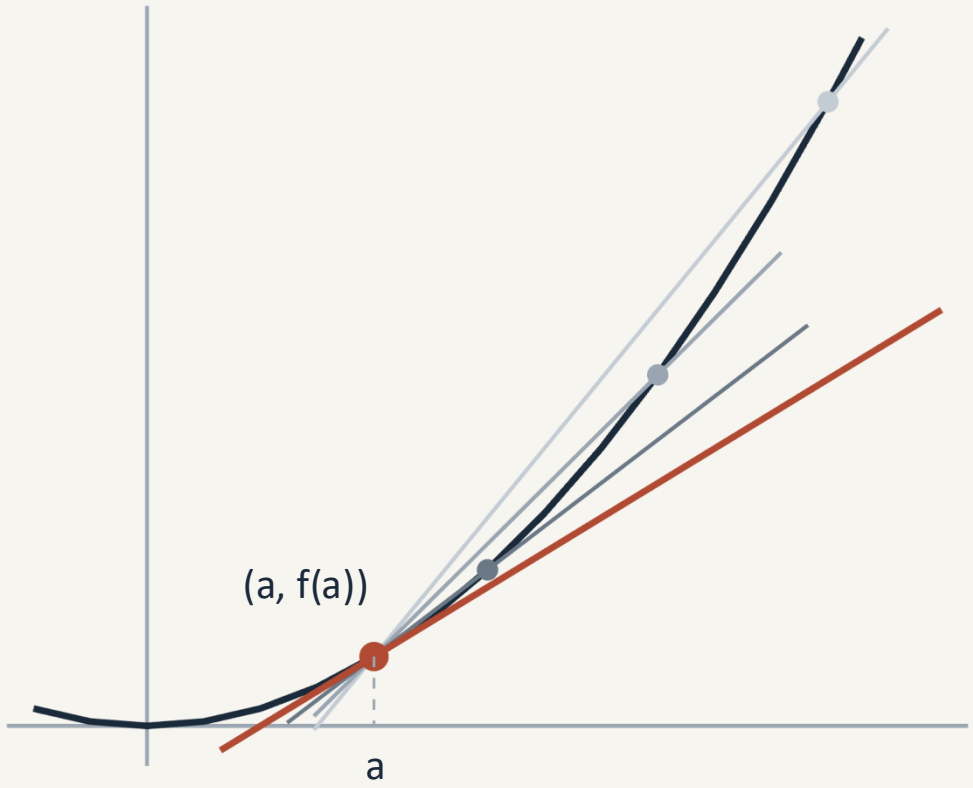
As h → 0, the second point slides toward (a, f(a)) and the secant lines approach the tangent line.

The slope of the tangent is the **instantaneous rate of change** of f at a:

$$m_{tan} = \lim_{h \to 0} \frac{f(a + h) - f(a)}{h}$$

secant lines, h getting smaller

tangent line at a



EXAMPLE 1

The tangent line to $f(x) = x^2$ at $x = 3$

1. Find the slope

$$f'(x) = \lim_{h \to 0} \frac{(x + h)^2 - x^2}{h}$$

$$= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{h}$$

$$= \lim_{h \to 0} (2x + h) = 2x$$

At $x = 3$: $f'(3) = 2 \cdot 3 = 6$

2. Find the line

Point: $(3, f(3)) = (3, 9)$
Slope: $m = 6$

Substitute into $y = mx + b$:

$$9 = 6 \cdot 3 + b$$

$$b = -9$$

$y = 6x - 9$

LINEARIZATION

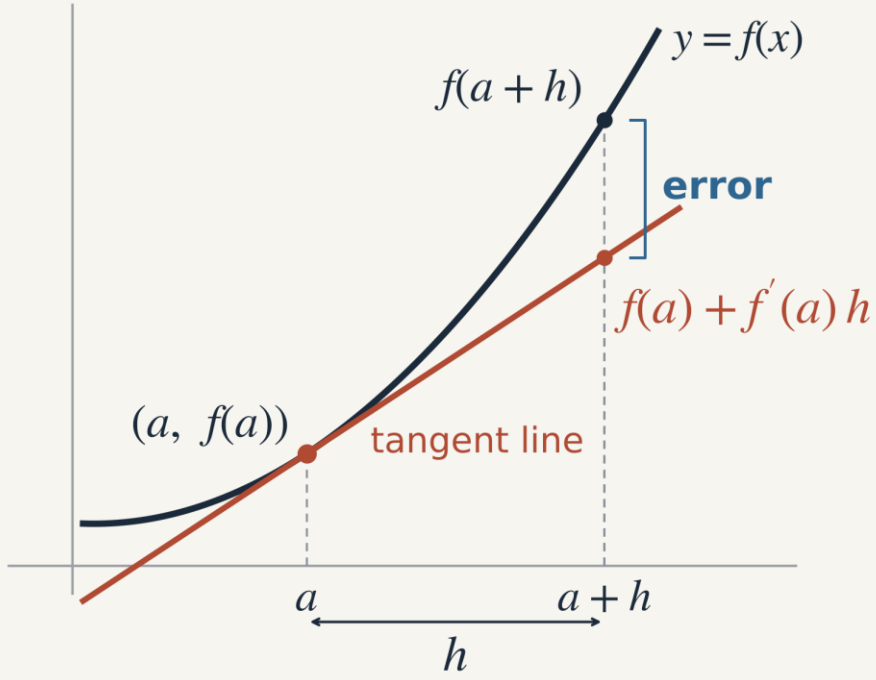
Near a point, a curve looks like its tangent line

Zoom in on a smooth curve and it straightens out. Near x = a, the tangent line is the **best straight-line approximation** of f.

$$f(a + h) \approx f(a) + f'(a) \cdot h$$

new value ≈ old value + slope × step

Example: f(x) = x² at a = 3 gives f(3.1) ≈ 9 + 6 · 0.1 = 9.6.
The true value is **9.61**, so the error is only 0.01.



Why AI cares

Step against the slope, h = −α f′(a), and f changes by about **−α f′(a)² ≤ 0**. A small step against the derivative always goes downhill. That is gradient descent.

DEFINITION

The derivative

At a point a

Let f be defined on an open interval containing a. The derivative of f at a is

$$f'(a) = \lim_{h \to 0} \frac{f(a + h) - f(a)}{h}$$

f is **differentiable at a** if this limit exists.

As a function of x

The derivative function f′ assigns to each x the value

$$f'(x) = \lim_{h \to 0} \frac{f(x + h) - f(x)}{h}$$

Its domain is every x where the limit exists.

NOTATION

Four ways to write the derivative of y = f(x)

$f'(x)$	$\frac{dy}{dx}$	y'	$\frac{d}{dx} f(x)$
Prime (Lagrange)	**Leibniz**	**Prime shorthand**	**Operator form**

All four mean the same thing. Leibniz's dy/dx is a useful reminder: a small change in y divided by a small change in x.

EXAMPLE 2

The derivative of f(x) = 3x + 1

$$f'(x) = \lim_{h \to 0} \frac{(3(x + h) + 1) - (3x + 1)}{h}$$

$$= \lim_{h \to 0} \frac{3h}{h} = 3$$

Takeaway

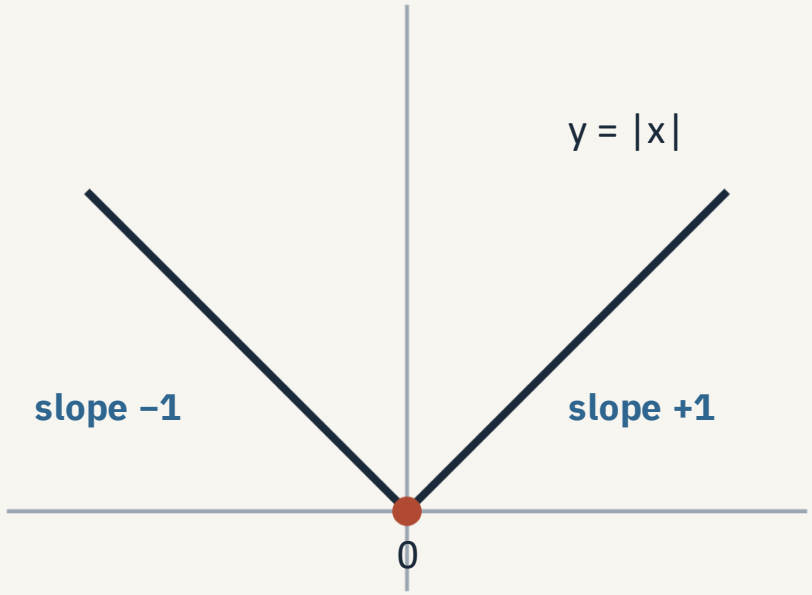
A line has the same slope everywhere, so its derivative is a constant: the slope of the line.

DIFFERENTIABILITY

When the derivative does not exist

$f(x) = |x|$ is continuous at 0, but its graph has a corner there.

- From the right, the slope is +1.
- From the left, the slope is −1.
- The one-sided limits of $|h|/h$ disagree, so $f'(0)$ does not exist.



differentiable ⇒ continuous

The reverse is false, and $|x|$ is the counterexample.

In AI: ReLU(x) = max(0, x), the most common activation function in neural networks, has the same corner at 0.

PART 2

Differentiation rules

Computing a limit every time is slow. Each of these rules is proved once from the definition, and then lets us differentiate almost any formula in a few lines.

RULE 1

Constant rule

If $f(x) = c$, then $f'(x) = 0$.

Why

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \to 0} \frac{c - c}{h} = 0$$

The graph of a constant function is a horizontal line, and a horizontal line has slope 0 everywhere.

RULE 2

Power rule

If $f(x) = x^n$, then $f'(x) = n\,x^{n-1}$.

Works for every real exponent n.

n = 3

$x^3 \to 3x^2$

n = 100

$x^{100} \to 100x^{99}$

n = −2

$x^{-2} \to -2x^{-3}$

n = ½

$\sqrt{x} \to 1/(2\sqrt{x})$

RULES 3–5

Sum, difference and constant multiple

Sum rule

$$\frac{d}{dx}[f(x) + g(x)]$$

$$= f'(x) + g'(x)$$

Differentiate term by term.

Difference rule

$$\frac{d}{dx}[f(x) - g(x)]$$

$$= f'(x) - g'(x)$$

The same, with a minus sign.

Constant multiple

$$\frac{d}{dx}[c \cdot f(x)]$$

$$= c \cdot f'(x)$$

Constants factor out.

EXAMPLE 3

The derivative of $f(x) = 2x^5 + 7$

$$f'(x) = \frac{d}{dx}(2x^5 + 7)$$

$$= \frac{d}{dx}(2x^5) + \frac{d}{dx}(7)$$ Sum rule

$$= 2 \cdot \frac{d}{dx}(x^5) + 0$$ Constant multiple and constant rules

$$= 2 \cdot 5x^4 = 10x^4$$ Power rule

RULE 6

Product rule

$$\frac{d}{dx}[f(x)\,g(x)] = f'(x)\,g(x) + f(x)\,g'(x)$$

Derivative of the first times the second, plus the first times the derivative of the second. f and g must both be differentiable.

Example: f(x) = 3x + 1 and g(x) = x²

$$\frac{d}{dx}[(3x+1)\,x^2] = 3 \cdot x^2 + (3x+1) \cdot 2x = 3x^2 + 6x^2 + 2x = 9x^2 + 2x$$

Check: expanding first gives $3x^3 + x^2$, whose derivative is also $9x^2 + 2x$.

RULE 7

Quotient rule

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x)\,f'(x) - f(x)\,g'(x)}{[g(x)]^2}$$

Memory aid: "low d-high minus high d-low, over low squared." Order matters because of the minus sign.

Tip: you can also write f / g as $f \cdot g^{-1}$ and use the product and chain rules. That is one less formula to remember.

TRIGONOMETRIC FUNCTIONS

Two derivatives to know

f(x)	f′(x)
sin x	cos x
cos x	−sin x

Why only two? sin and cos appear in ML, for example in the positional encodings of transformers. tan, cot, sec and csc rarely do.

Example: $f(x) = \cos x + \sin x \Rightarrow f'(x) = -\sin x + \cos x$

RULE 8

Chain rule

If $h(x) = f(g(x))$, then $h'(x) = f'(g(x)) \cdot g'(x)$.

Derivative of the outside, with the inside left alone, times the derivative of the inside.

$$h(x) = \frac{1}{(3x^2 + 1)^2} = (3x^2 + 1)^{-2}$$

$$h'(x) = -2(3x^2 + 1)^{-3} \cdot 6x = \frac{-12x}{(3x^2 + 1)^3}$$

Outside: $f(u) = u^{-2}$, so $f'(u) = -2u^{-3}$

Inside: $u = g(x) = 3x^2 + 1$, so $g'(x) = 6x$

EXPONENTIAL FUNCTIONS

e^x is its own derivative

$$\frac{d}{dx}(e^x) = e^x$$ Natural exponential

$$\frac{d}{dx}(e^{g(x)}) = e^{g(x)} \cdot g'(x)$$ With the chain rule

$$\frac{d}{dx}(b^{g(x)}) = b^{g(x)} \cdot g'(x) \cdot \ln b$$ Any base b > 0

Example: $f(x) = e^{\sin(2x)}$

$f'(x) = e^{\sin(2x)} \cdot \cos(2x) \cdot 2 = 2\cos(2x)\, e^{\sin(2x)}$

THE SIGMOID FUNCTION

The sigmoid turns any number into a probability

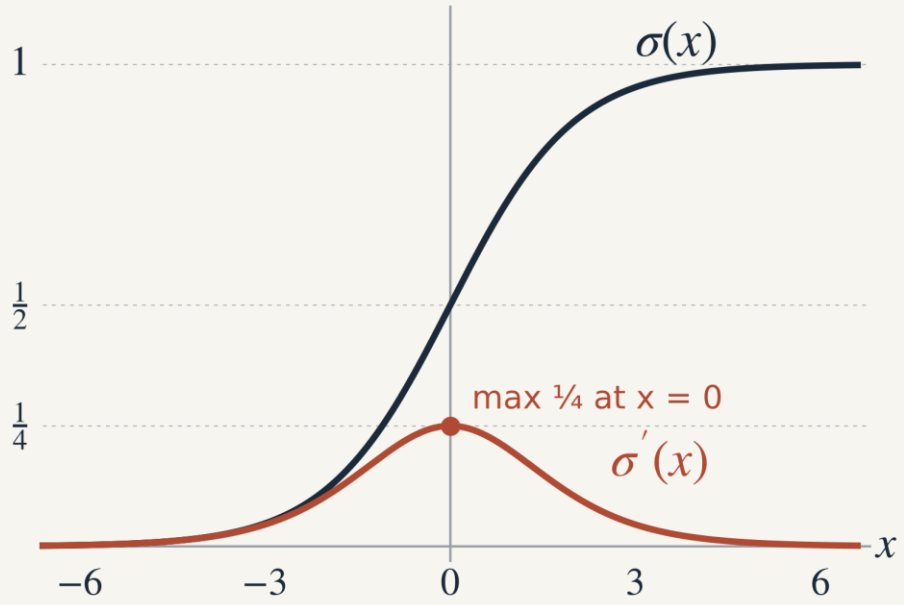
σ squashes every real number into (0, 1), so its output can be read as a probability. Its derivative is a chain-rule exercise.

$$\sigma(x) = 1 / (1 + e^{-x})$$

$\sigma(x) = (1 + e^{-x})^{-1}$

$\sigma'(x) = -(1 + e^{-x})^{-2} \cdot (-e^{-x}) = e^{-x} / (1 + e^{-x})^2$

$\sigma'(x) = \sigma(x) \cdot (1 - \sigma(x))$



Why AI cares

σ′ reuses σ, which the network has already computed, so it is almost free. And **σ′ ≤ ¼**: multiply many of these together and the gradient vanishes (will see in a future lecture).

LOGARITHMIC FUNCTIONS

The derivative of ln x is 1/x

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

For x > 0

$$\frac{d}{dx}\ln(g(x)) = \frac{g'(x)}{g(x)}$$

With the chain rule

$$\frac{d}{dx}\log_b(g(x)) = \frac{g'(x)}{g(x) \cdot \ln b}$$

Any base b > 0, b ≠ 1

Example: the derivative of $\ln(x^2 + 1)$ is $2x / (x^2 + 1)$.

LOGARITHMS IN ML

Logs turn products into sums

A likelihood is a product of probabilities, one per data point. The log turns it into a sum, which is easy to differentiate.

$$\ln(p_1\, p_2 \cdots p_n) = \ln p_1 + \ln p_2 + \cdots + \ln p_n$$

Example: heads has probability p, and we see H, H, T.

$L(p) = p \cdot p \cdot (1 - p) = p^2(1 - p)$

$\ln L = 2 \ln p + \ln(1 - p)$

$d/dp \ln L = 2/p - 1/(1 - p) = 0 \;\Rightarrow\; \mathbf{p = 2/3}$

Why computers need it

1000 probabilities of 0.1, multiplied

0

10^{-1000} is too small, so it rounds to 0

The same 1000 logs, added

−2302.6

an ordinary number, safe to work with

This is why ML code uses **log-probabilities**.

HIGHER-ORDER DERIVATIVES

Differentiate again

The second derivative f'' is the derivative of f', the third derivative f''' is the derivative of f'', and so on. In Leibniz notation the second derivative is d^2y/dx^2.

POSITION

$s(t) = t^3 - 6t^2$

→

VELOCITY

$v(t) = s'(t)$
$= 3t^2 - 12t$

→

ACCELERATION

$a(t) = s''(t)$
$= 6t - 12$

At $t = 1$: $v = -9$ and $a = -6$. Both are negative, so the object is moving backward and speeding up.

REFERENCE

The rules at a glance

Rule	f(x)	f′(x)
Constant	c	0
Power	x^n	$n\,x^{n-1}$
Sum, difference	$f \pm g$	$f' \pm g'$
Constant multiple	$c\,f$	$c\,f'$
Product	$f\,g$	$f'\,g + f\,g'$

Rule	f(x)	f′(x)
Quotient	f / g	$(g\,f' - f\,g') / g^2$
Chain	$f(g(x))$	$f'(g(x)) \cdot g'(x)$
Exponential	e^x, b^x	e^x, $b^x \ln b$
Logarithm	$\ln x$	$1 / x$
Trigonometric	$\sin x$, $\cos x$	$\cos x$, $-\sin x$
Sigmoid	$\sigma(x)$	$\sigma(x)(1 - \sigma(x))$

PRACTICE

Your turn: find each derivative

(a) · Hint: chain rule

$$h(x) = \cos(5x^2)$$

(b) · Hint: product rule, then chain rule

$$h(x) = (2x + 1)^5 \, (3x - 2)^7$$

(c) · Hint: chain rule, and $\sigma' = \sigma(1 - \sigma)$

$$h(x) = \sigma(3x + 1)$$

(d) · Data H, T, T. Also find where it is zero

$$\frac{d}{dp} \left[\ln p + 2 \ln(1 - p) \right]$$

PRACTICE

Answers

(a) $h(x) = \cos(5x^2)$

$h'(x) = -10x \sin(5x^2)$

(b) $h(x) = (2x + 1)^5 (3x - 2)^7$

$h'(x) = 10(2x + 1)^4(3x - 2)^7 + 21(2x + 1)^5(3x - 2)^6$

(c) $h(x) = \sigma(3x + 1)$

$h'(x) = 3\,\sigma(3x + 1)(1 - \sigma(3x + 1))$

(d) $\ln L = \ln p + 2 \ln(1 - p)$

$1/p - 2/(1 - p)$, which is zero at $p = 1/3$

PART 3

Numerical differentiation

When f is known only through measurements, estimate $f'(a)$ from nearby values. With a step size h, three approximations are:

Forward difference

$$\frac{f(a+h)-f(a)}{h}$$

Error shrinks like h

Backward difference

$$\frac{f(a)-f(a-h)}{h}$$

Error shrinks like h

Central difference

$$\frac{f(a+h)-f(a-h)}{2h}$$

Error shrinks like h^2, usually far more accurate

EXAMPLE 4

How fast is the runner going?

t (s)	0.0	0.5	1.0	1.5	2.0
x (m)	0.00	3.65	6.80	9.00	12.15

$$v(0.5) \approx \frac{x(1.0) - x(0.0)}{2 \cdot 0.5} = \frac{6.80 - 0.00}{1.0} = 6.8 \text{ m/s}$$

h = 0.5

$$v(1.0) \approx \frac{x(1.5) - x(0.5)}{2 \cdot 0.5} = \frac{9.00 - 3.65}{1.0} = 5.35 \text{ m/s}$$

h = 0.5 · the opening question

$$v(1.25) \approx \frac{x(1.5) - x(1.0)}{2 \cdot 0.25} = \frac{9.00 - 6.80}{0.5} = 4.4 \text{ m/s}$$

h = 0.25 · uses data at 1.0 and 1.5

ACCURACY

Three estimates of the velocity at t = 0.5 s

Method	Calculation	Estimate
Forward	(6.80 − 3.65) / 0.5	6.3 m/s
Backward	(3.65 − 0.00) / 0.5	7.3 m/s
Central	(6.80 − 0.00) / 1.0	6.8 m/s

Why central wins

The forward and backward estimates err in opposite directions. The central difference is their average, (6.3 + 7.3) / 2 = 6.8, so much of the error cancels.

In general, halving h roughly halves the error of a one-sided difference but cuts the error of the central difference to about a quarter.

GRADIENT CHECKING

Use finite differences to check a derivative

Worked out a derivative by hand, or coded one? Compare it with the central difference at a small step, such as $h = 10^{-5}$.

$$f'(a) \approx [f(a + h) - f(a - h)] / 2h$$

If the two agree to many digits, the derivative is right. If they disagree, **there is a bug**.

Why AI cares

This is the standard test for backpropagation code. We use it again in the future.

Example: σ′(2)

Formula $\sigma(2) \cdot (1 - \sigma(2))$	**0.1049936**
Central difference $h = 10^{-5}$	**0.1049936**
Buggy formula $\sigma(2) \cdot (1 + \sigma(2))$	**1.6566**

Rows 1 and 2 agree to 10 digits. Row 3 does not, so its formula is wrong.

WRAP-UP

Key takeaways

The idea

The derivative is the limit of secant slopes: the slope of the tangent line and the instantaneous rate of change.

The definition

$f'(x) = \lim_{h \to 0} [f(x + h) - f(x)] / h$. Near a point, f is close to its tangent line.

The rules

Power, product, quotient and chain rules handle almost any formula, including $\sigma' = \sigma(1 - \sigma)$. Logs turn products into sums.

From data

Use the central difference $[f(a + h) - f(a - h)] / 2h$ to estimate a rate, or to check a derivative you computed.

Thank you. Questions?